

Individual Contributions

Group 04 - MicroInspect

The specific technical contributions and responsibilities of each team member for the PCB Bare-Board Defect Detection project are summarized below:

- **E/23/336 - S.M.D.S.B. Samarakoon**

- Developed the classical topological classification algorithm.
- Conducted experiments on how the sensor noise in images affect the baseline method performance.
- Handled the core dataset preprocessing (image tiling and overlapping).
- Created and deployed training environment in Ada (ada.ce.pdn.ac.lk) server to train heavy weight deep learning models.
- Trained both the YOLOv11-Nano and YOLOv11-Medium deep learning models.
- Experimented on YOLOv11-Medium model by fine-tuning parameters to achieve the highest possible precision and recall metrics.
- Implemented Test-Time Augmentation (TTA) in the core YOLO inference pipeline.

- **E/23/104 - G.P.M. Gamage**

- Researched and experimented with different image processing techniques for PCB image registration and enhancement.
- Developed the baseline classical defect detection method, establishing the foundational algorithmic pipeline for the project.
- Led the development of the user interface.
- Built and refined the web dashboard (demo UI) that allows users to upload images and compare the classical and YOLO model outputs side-by-side.

- **E/23/035 - P.I.N. Bandara**

- Developed the evaluation scripts to calculate mAP, Precision, and Recall.
- Handled the overall repository cleanup, structural refactoring, and reproducibility overhauls.
- Documentation - The full README (structure, setup, dataset download commands, usage workflows, weight download links)
- Bug Fixes - Fixed output directory path resolutions, Replaced hardcoded Linux absolute paths

- Evaluation Script Updates - Refactored scripts/evaluate.py: updated evaluation metrics logic.
- **E/23/117 - W.R.A.D.N. Gunathilake**
 - Managed the initial project environments and repository set-up.
 - Assisted with the literature review on classical AOI systems.
 - Researched for different approaches to implement computer vision models for microscopic defect detection.
 - Assisted with data pre processing and Deep learning method training on GPU server and experimentations with YOLOv11.
 - Handled the final report formatting and documentation structure.

Group 4:

E/23/035 - P.I.N. Bandara	- e23035@eng.pdn.ac.lk
E/23/104 - G.P.M. Gamage	- e23104@eng.pdn.ac.lk
E/23/117 - W.R.A.D.N. Gunathilake	- e23117@eng.pdn.ac.lk
E/23/336 - S.M.D.S.B. Samarakoon	- e23336@eng.pdn.ac.lk